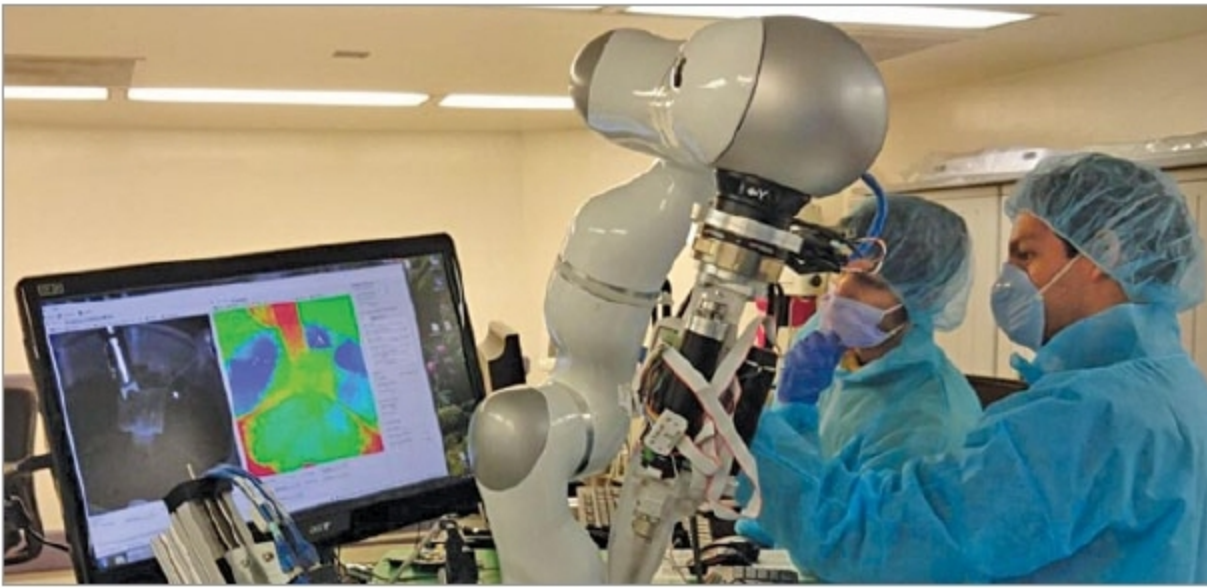




An autonomous robot surgery (Credit: <https://blogs.nvidia.com>)

cate or otherwise respond to them as per input data. For example, in image processing, input data is fed from vision sensors in a camera system. No matter how complex or good the algorithm is, without a good vision sensor the output will not be accurate as expected.

**Processors.** The central processing unit (CPU) and graphics processing unit (GPU) are equally important in AI technology. GPU is designed to manipulate and alter memory to accelerate the creation of images. A cluster of GPUs is required to train a model with a training data set consisting of thousands of images. Depending on data size and performance of the model, the number of input images and SLA expected, a cluster of processors are used to meet application requirements.

**Storage.** To train AI and machine learning algorithms, large datasets are needed to deliver accurate decisions. This means that the demand for data storage space is significant. Managing data sets requires storage systems that can scale without limits. Your data set may not fit on the disk of a laptop, and you may need a number of SSDs and/or HDDs to store the ever-increasing training data.

## AI helps make better and smarter decisions

AI technology is employed in safety-critical situations such as airports, ATM machines and aircraft operations. When decision-makers and business executives have reliable data analyses, recommendations and follow-ups through AI systems, they can make smarter decisions and better

choices for their business, employees and future actions.

**Automotive.** In the automotive industry, several AI applications have been developed from vehicle design to marketing and sales decision-making support. For instance, AI has led to the design of smarter cars, equipped with multiple sensors that learn and identify patterns, add-on safe-drive features that warn drivers of possible collisions and lane departures. AI is used to map the decisions made by car buyers and manufacturers, and transportation services providers.

**Self-driving cars.** AI takes decisions by itself in a self-driving car where computer vision and machine learning plays a vital role. There are a number of interesting and challenging decision-making problems starting from autonomous driving of a single vehicle to coordination among multiple vehicles. AI algorithms take care of these problems through an autonomous system throughout the course of driving till the destination is reached.

**Financial.** In financial services, the model is designed to help financial service companies map buyer personas and anticipate customer behaviour. AI-enabled financial service companies can validate real-time business decisions within seconds.

**Railway system.** Use of AI within the rail sector is critical to making the rail experience more intuitive, and to eliminate many of the problems faced by passengers. Computational intelligence (CI) is used to forecast problems in railway maintenance, which is important and critical for safe, comfortable and economic transportation.

Before making any decision, rea-

soning is an essential step to explore the links between AI and decision-making. There are two aspects of decision-making: diagnosis and look-ahead. AI has many relationships with diagnosis involving expert systems, case-based reasoning, fuzzy set and rough set theories; and look-ahead reasoning involving uncertainty and preferences.

Traditionally, human operators have had to rely on inconsistent and incomplete data. With AI, data-based models and simulations are now being used to solve various problems.

**Healthcare.** AI is critical here because less accurate or unreliable results will put patients at risk. Therefore AI has to be accurate, and safe AI programs need to be trained, or at least constantly updated, to be able to identify new and exceptional health cases.

Remote AI-assisted robotic surgeries have been successfully carried out by many experts across the globe. With technology advancements, AI has helped personalised healthcare services, reduced the time to look for information that is critical to decision-making and facilitated better care for patients.

One of the more critical applications of AI in healthcare is ensuring security of data and solutions.

## Conclusion

A few cases of AI in mission-critical applications including railway systems, healthcare and self-driving cars have been covered. While we are smarter and more intelligent than present machines, AI technology certainly helps us in the decision-making process to achieve better outcome. Mission-critical applications require smarter decisions, better accuracies and results. **Although AI machines are used in many applications, very critical applications require the interventions of human operators.**

In mission-critical AI applications, both hardware and software are equally important. Key features in mission-critical AI applications include accuracy, scale, strict SLAs and security. AI plays an important role in decision-making in various industries including railway systems, healthcare, self-driving, businesses and others. **EFY**